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$$I = \int \frac{4dx}{2x^2 + 1}$$

$$\text{Stel: } t = \sqrt{2}x \Rightarrow dt = \sqrt{2}dx \Leftrightarrow dx = \frac{dt}{\sqrt{2}}$$

$$= \int \frac{4}{t^2 + 1} \frac{dt}{\sqrt{2}} = 4 \operatorname{Bgtan}(t) + c$$

$$= \frac{4}{\sqrt{2}} \operatorname{Bgtan}(\sqrt{2}x) + c = 2\sqrt{2} \operatorname{Bgtan}(\sqrt{2}x) + c$$

References